



Cross-Species Self-supervised Transfer Learning for Pulmonary Lobe Segmentation in Nonhuman Primates

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Abstract

Annotations of 3D medical images for segmentation require specialized expertise and are time-consuming, making large, labeled datasets rare and challenging to produce. Our objective was to investigate whether large unlabeled human datasets can be leveraged using cross-species self-supervised transfer learning to enhance the segmentation of pulmonary lobes in computed tomography (CT) scans from nonhuman primates with and without lower respiratory infection. A total of 1667 unlabeled human chest CT scans were assembled from two publicly available sources, and 23 chest CT scans of crab-eating macaques were annotated for the locations of the pulmonary lobes. The unlabeled human scans were used to train a 3D vision transformer (ViT) autoencoder in a self-supervised manner using contrastive learning. The pretrained ViT encoder was transferred to a U-Net transformers (UNETR) segmentation model, which was then trained using the labeled macaque dataset to perform pulmonary lobe segmentation. Ablation experiments on the effects of self-supervised pretraining, layer freezing, and data augmentation were conducted. The segmentation model with cross-species self-supervised pretraining achieved high performance (Dice similarity coefficient (DSC) = 90.31 ± 1.77) that was significantly greater than without pretraining ($\Delta\text{DSC} = 1.2\%$, $t_{\text{paired}} = 5.3$, $p = 1.8\text{E-}3$). Ablation experiments on the human pretraining data demonstrated that the amount of data and diversity of sources were important to performance. In fine-tuning, freezing just the first three layers of the ViT produced the best-performing model, and data augmentation before self-supervised pretraining and supervised fine-tuning was critical for high performance. Cross-species self-supervised transfer learning significantly improved the macaque pulmonary lobe segmentation performance with no additional acquisition or annotation costs.

Keywords Self-supervised learning · Transfer learning · Cross-species · Segmentation · Computed tomography · Vision transformer

Abbreviations

2D	Two-dimensional	IRF-Frederick	Integrated Research Facility at Fort Detrick
3D	Three-dimensional	MLP	Multilayer perceptron
AI	Artificial intelligence	MONAI	Medical open network for AI
ASSD	Average symmetrical surface distance	NIAID	National Institute of Allergy and Infectious Diseases
CNN	Convolutional neural network	OCICB	Office of Cyber Infrastructure and Computational Biology
COVID-19	Coronavirus disease 2019	SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
CT	Computed tomography	TCIA	The Cancer Imaging Archive
DSC	Dice similarity coefficient	UNETR	U-Net transformers
GPU	Graphics processing unit	ViT	Vision transformer
HD ₉₅	Hausdorff distance 95th percentile		
HPC	High performance computing (cluster)		
IoU	Intersection over Union		

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Introduction

Data scarcity is a core challenge of training deep learning models for semantic segmentation of 3D medical images. Labeled 3D medical imaging datasets are rare and expensive because their creation is time-consuming and requires specialized expertise. For example, to generate segmentation annotations on high-resolution 3D computed tomography (CT) images, radiologists must draw 2D segmentations on each of the approximately 200–300 slices of the third dimension. Although animal models are a cornerstone of biomedical research, labeled imaging data are scarce even for commonly used animals (e.g., laboratory mice or rats) and are rare or virtually nonexistent for less commonly used models, such as nonhuman primates, domestic ferrets, and domesticated guinea pigs [1]. With particular regard to novel or (re)emerging infectious diseases in humans, a diversity of animal models may be needed to best model infection and/or disease, depending on the pathogen (e.g., macaques and domestic ferrets as models of Ebola virus disease [2]). The potential for AI-enabled analysis of medical imaging data to characterize disease and explore pathogenesis in humans and animal models has been newly recognized [1, 3–6]. However, the scarcity of labeled data presents a major challenge to forward progress and future agility in deploying AI-enabled imaging research tools.

Of many deep learning approaches that have been proposed to maximize model performance with a limited labeled dataset, *transfer learning* is particularly effective at facilitating *few-shot learning* (i.e., learning with a small amount of labeled data). Transfer learning is the process of pretraining a model on a related task and then fine-tuning the model to the target task. *Cross-species* transfer learning uses data from animals of one species to pretrain a model to perform a task in those of another species. Cross-species transfer learning may enable few-shot learning when similar images have been collected but in asymmetric quantities, including when human infections are relatively rare (i.e., more data have been collected on nonhuman animals than humans) or when animal model data are relatively rare (i.e., more data have been collected on humans than the target nonhuman animal). Despite the potential performance benefits, the effectiveness and best strategies for cross-species transfer learning for this purpose have been under-explored.

Cross-species transfer learning has been examined for human-to-nonhuman-animal [7–10], nonhuman-animal-to-human [11], and nonhuman-animal-to-nonhuman-animal [12] transfer learning, but only with labeled datasets for *supervised* pretraining. In *self-supervised* pretraining, the model is trained on tasks that do not require labeled data,

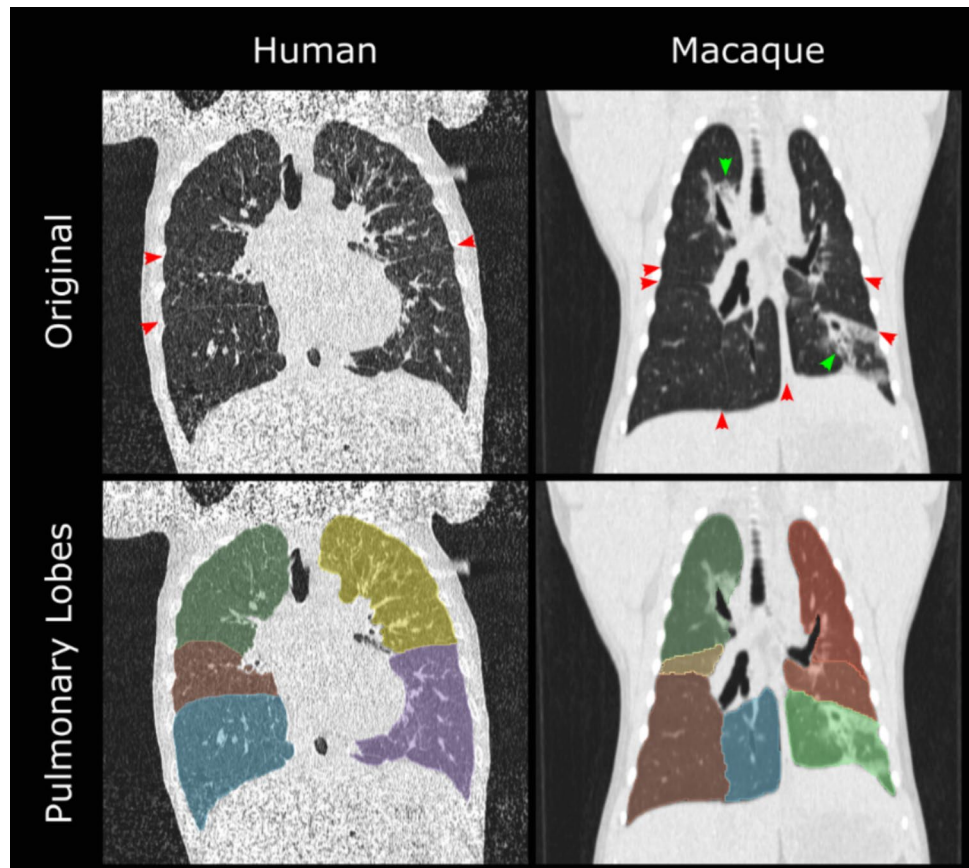
thereby avoiding the immense cost of generating a labeled pretraining dataset. The major challenge of this approach is that the pretraining task is typically less relevant to the fine-tuning task, making the selection of the pretraining dataset particularly important. To our knowledge, no study has examined the use of self-supervised cross-species transfer learning for image segmentation.

The spatial distribution of pulmonary lesions varies across respiratory infectious diseases in a manner that informs diagnosis, classification, and outcome prognostication. Disease characterization on a per-lobe basis is clinically and scientifically useful, but accurate pulmonary lobe segmentation to enable this anatomic characterization is challenging, even when annotated manually by human radiologists. For one, pulmonary fissures that delineate each lung lobe are thin, often only visible on limited imaging slices, and may be obscured by or indistinguishable from other radiodense lung structures, including blood vessels and disease abnormalities. In fact, lung abnormalities typical of lower respiratory diseases (e.g., ground-glass opacities, consolidation, pleural effusion) are likely to distort pulmonary lobe boundaries. Anatomic differences between models of different animal species further add to the challenge: for example, human pulmonary lobe segmentation models are ineffective for segmentation of crab-eating macaque lobes because crab-eating macaque lungs are comprised of seven lobes whereas human lungs have five (Fig. 1). Therefore, specialized models are needed for deep-learning-based pulmonary lobe segmentation for macaque lungs.

The U-Net transformers (UNETR) architecture utilizes a U-shaped encoder–decoder architecture but with a transformer encoder to more effectively capture global multi-scale information [13]. Global context is particularly important for pulmonary lobe segmentation given the importance of each lobe’s location relative to the body and the lack of local patterns/intensities that are specific to a particular lobe. One limitation of transformer architectures is that they often have more trainable parameters and therefore require more data compared to fully convolutional neural networks (CNNs). We hypothesized that this limitation may be overcome by self-supervised pretraining to maintain the global context benefits of transformer architectures despite limited labeled data.

In this study, we explored cross-species self-supervised transfer learning for pulmonary lobe segmentation of crab-eating macaques with and without lower respiratory infection. We utilized self-supervised learning on a large unlabeled human chest CT dataset to pretrain a vision transformer (ViT) autoencoder model. The ViT block was used as the encoder for a UNETR model. We manually annotated the boundaries of pulmonary lobes on chest CT scans of macaques and used these labeled data to fine-tune the UNETR model to automatically segment the pulmonary

Fig. 1 Human and crab-eating macaque lung anatomy. Computed tomography (CT) images of human and crab-eating macaque lungs are shown in the coronal plane. Humans have five pulmonary lobes (two left and three right), whereas crab-eating macaques have seven pulmonary lobes (three left, three right, and one auxiliary). Vessels, airways, and disease lesions can obstruct faint pulmonary fissures that show the boundaries between lobes. Red arrowheads point to the locations of the pulmonary fissures, and green arrowheads point to pulmonary lesions. Window = 1400 HU, level = -500 HU



lobes. The average performance of the model on all lobes as well as the performance of the model for each individual lobe was evaluated. Ablation experiments were performed to quantify the effects of (1) self-supervised pretraining, (2) layer freezing, and (3) data augmentation. To our knowledge, this study is the first to explore cross-species self-supervised transfer learning for image segmentation.

Materials and Methods

Datasets

This study utilized a total of 1690 3D CT scans consisting of 1667 unlabeled human chest CT scans from public sources [14, 15] and 23 chest CT scans of crab-eating macaques (*Macaca fascicularis* Raffles, 1821) acquired and manually annotated for the locations of the seven pulmonary lobes at the National Institute of Allergy and Infectious Diseases (NIAID) Integrated Research Facility at Fort Detrick (IRF-Frederick). The 23 chest CT scans from 12 macaques (one to four scans per macaque) were aggregated from multiple imaging studies performed at the IRF-Frederick and included CT scans from macaques at various time points after severe acute respiratory syndrome

coronavirus 2 (SARS-CoV-2) exposure as well as from unexposed macaques (ratio of 17:6 exposed to unexposed). Using images from a related domain improves pretraining performance [16]. Pretraining datasets were chosen with the goal of teaching the model to learn three key features from the CT scan: (1) the typical distribution of intensities, (2) the general shapes and spatial structures, and (3) the variability that can result from a lower respiratory infection. Therefore, two human CT datasets were used in this study: (1) The Cancer Imaging Archive (TCIA) “CT Images in COVID-19” dataset [14], henceforth referred to as “Dataset T,” and (2) the Harvard Dataverse “COVID-19-CT-Dataset” [15], henceforth referred to as “Dataset H.” Dataset T consisted of 771 human chest CT scans from 661 patients with SARS-CoV-2 infection confirmed by reverse transcription polymerase chain reaction. Dataset H consisted of 923 human chest CT scans from an equal number of patients with SARS-CoV-2 infection confirmed by reverse transcription polymerase chain reaction. A total of 27 scans were excluded from Dataset H after manual screening for poor image quality, resulting in a total of 896 scans from Dataset H. As a result, the combined human dataset consisted of 1667 scans from 1557 patients. Additional details on the datasets can be found in the associated publications for Dataset T [17, 18] and Dataset H [19].

Macaque CT Acquisition

Macaque chest CT images were acquired on a Philips Gemini 16-slice CT scanner (Philips Healthcare, Cleveland, OH, USA) in helical scan mode. The scanner parameters were set at 140 kVp, 300 mA per slice, 1-mm thickness, 0.5-mm increment, 0.688-mm pitch, 16×0.75 collimation, and 0.75-s rotation. CT image reconstruction used a 512×512 matrix for a 250-mm transverse field of view (pixel size = 0.488 mm). During acquisition, each macaque was subject to a breath-hold of 15–20 s at a pressure of 150 mmH₂O.

Macaque CT Lung Lobe Annotation

Macaque chest CT scans were annotated for the locations of the seven pulmonary lobes in a semi-automated manner. A MIM (version 7.1.6, MIM Software, Cleveland, OH, USA) custom workflow was used to make an initial rough lobe annotation. Rough annotations were then corrected by two research trainees to more accurately follow the contours of the pulmonary interlobar fissures. Trainee annotations were then reviewed and further edited by two radiologists. Overall, the annotation process was difficult and time-consuming (approximately 32 h per CT scan) given that interlobar fissures were often faint and approximately 200 coronal slices needed to be annotated to cover the 3D extent of all seven pulmonary lobes.

Implementation Details

Our approach was implemented in Python v3.9.19, PyTorch v2.2.2, and medical open network for AI (MONAI) v1.3.0 [20]. The models were trained on NVIDIA A100 80GB graphics processing units (GPUs) housed at the Office of Cyber Infrastructure and Computational Biology (OCICB) High Performance Computing (HPC) cluster at the National Institute of Allergy and Infectious Diseases (NIAID), Bethesda, MD, USA.

Model Architecture

To examine the potential for cross-species transfer learning, a semi-supervised (using both labeled and unlabeled data) deep learning model was trained to perform semantic segmentation of the seven macaque pulmonary lobes from chest CT scans (Fig. 2) [21].

Preprocessing and Data Augmentation

Macaque CT scans were first cropped to $340 \times 340 \times 320$ from the center to remove parts of the scan outside the lung parenchyma. Human and macaque CT scans were both resampled to $192 \times 192 \times 192$ to produce a consistent matrix size that balanced segmentation resolution with memory constraints.

In both pretraining and fine-tuning stages, data augmentation of the training set was used to increase the variability of the input data and make the model more generalizable. The augmentations were designed to mimic the range of variability typically seen across scans, specifically differences in signal-to-noise and slight spatial variations (from changes in position and subject size/proportions). Two layers of augmentation were applied using MONAI transforms: first, either Gaussian noise ($\sigma = [0, 1]$), Gaussian smoothing ($\sigma = 0.25\text{--}1.5$), or neither was applied at a probability of 25, 25, and 50%, respectively; second, a 3D elastic deformation ($\sigma = 5\text{--}7$, magnitude = 50–150, rotation = $-\pi/2$ to $\pi/24$, scale = 0.95–1.05, translate = -50 to 50) was applied at a probability of 50%.

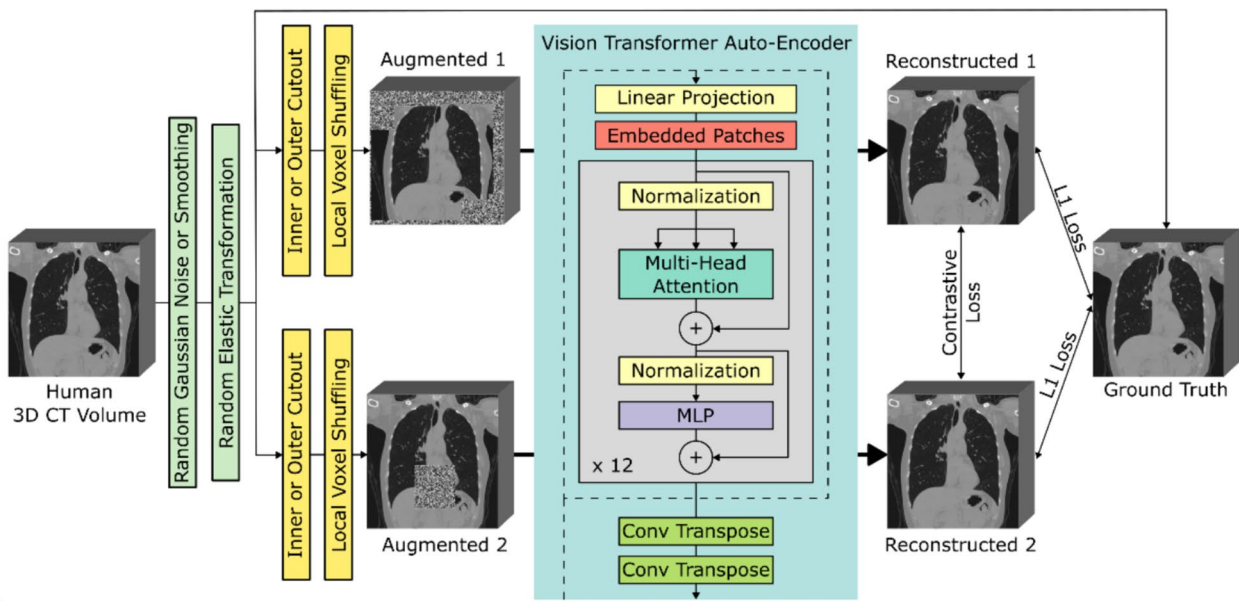
Self-supervised Pretraining

A 3D ViT autoencoder model was trained in a self-supervised (unsupervised) manner using contrastive learning [22] with unlabeled human chest CT scans. The ViT autoencoder consisted of the ViT encoder block followed by two convolutional transpose layers to generate a reconstruction similar to the size of the input image [21]. The goal of the pretraining was to train the ViT encoder for later use in a UNETR segmentation model. Before being fed to the ViT encoder, images were decomposed into a series of patch embeddings with position encodings. The ViT encoder consists of alternating multiheaded self-attention and multilayer perceptron (MLP) blocks [23]. Contrastive learning aims to learn representations by maximizing agreement between differently augmented views of the same data. For each image, two sets of random augmentations were applied, producing two augmented images. The random augmentations were inner or outer cutout [24, 25] and local voxel shuffling [26] for an inpainting pretext task. Outer and inner cutout augmentation involved the removal of random patches from both the edge and inside of the image; removed patches were then filled with random values within the range of the values of the original image. Local voxel shuffling augmentation is similar but maintains the intensity distribution of the original patch by shuffling the voxel values within random patches of the image. The ViT autoencoder was trained to reconstruct the original image from each augmented image while learning good representations using a combination of reconstruction and contrastive loss functions.

The contrastive loss function (i.e., NT-Xent; normalized temperature-scaled cross entropy [22]) is calculated between the reconstructed images, driving reconstructed images from the same source image together and reconstructed images from different source images apart:

$$l_{\text{contrastive}} = -\log \frac{\exp\left(\frac{\text{sim}(z_i, z_j)}{\tau}\right)}{\sum_k^{2N} 1_{k \neq i} \exp\left(\frac{\text{sim}(z_i, z_k)}{\tau}\right)} \quad (1)$$

a



b

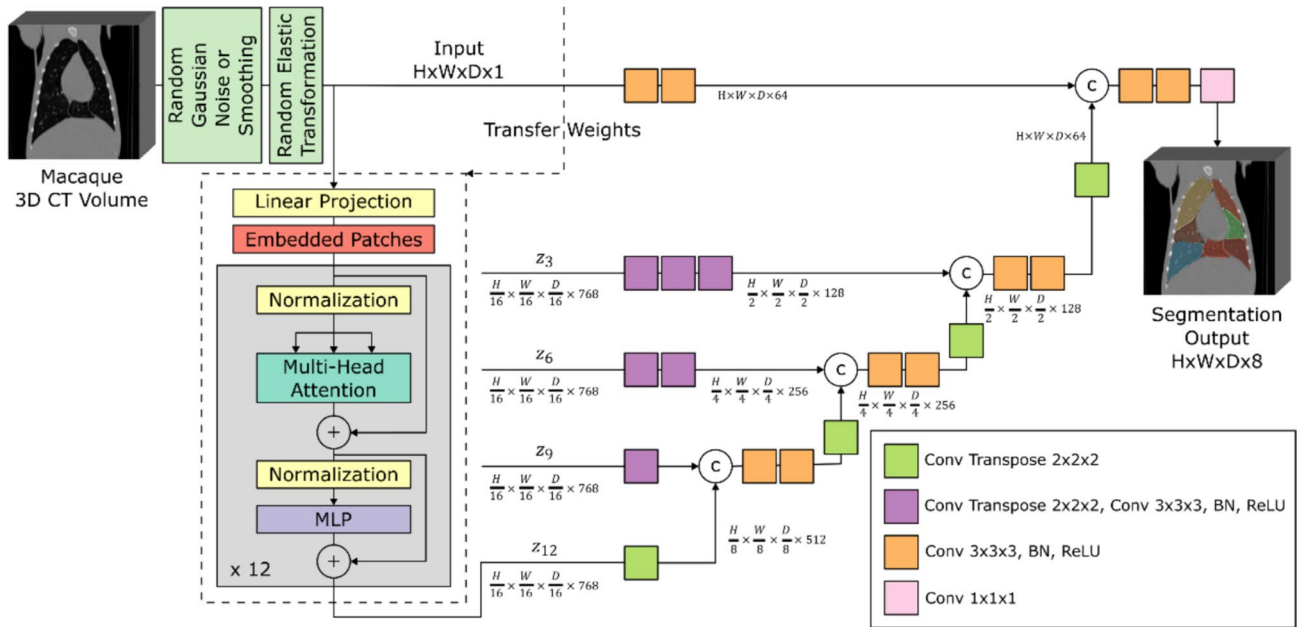


Fig. 2 Model architecture. **a** Self-supervised pretraining on human computed tomography (CT) data; a 3D vision transformer (ViT) autoencoder was trained using contrastive learning (self-supervised) on human CT data. **b** Supervised fine-tuning on macaque CT data; the transformer model weights were then used to initialize a 3D

U-Net transformers (UNETR) pulmonary lobe segmentation model that was trained using supervised learning on labeled macaque CT data. BN, batch normalization; Conv, convolution; D, depth; H, height; MLP, multilayer perceptron; ReLU, rectified linear unit; W, width

where $\text{sim}(x, y) = x^T y / \|x\| \|y\|$ denotes the cosine similarity between embeddings, 1 is an indicator function evaluating to 1 iff $k \neq i$, and τ denotes a temperature parameter.

The reconstruction loss is the mean absolute error (i.e., l_1 loss) between the reconstructed image and the original un-augmented image:

$$l_{reconstruction} = \text{mean}(\{l_1, \dots, l_n\}) \quad (2)$$

where $l_n = |x_n - y_n|$ for every reconstructed (y_n) and original (x_n) image pair.

The total loss combines the reconstruction and contrastive loss such that contrastive loss is prioritized when the reconstruction loss is high:

$$l_{total} = l_{reconstruction} + (l_{contrastive} \times l_{reconstruction}) \quad (3)$$

The training-validation data split ratio was 80:20. The validation loss was calculated as the reconstruction loss, and the model weights from the epoch with the lowest validation loss were saved for model fine-tuning. The model was pretrained for 350 epochs on 1667 3D CT images and took approximately 132 h on a platform with four central processing unit (CPU) cores and one GPU. The model was trained with a batch size of three, given GPU constraints. Model weight updates were performed using an Adam optimizer with a fixed learning rate of $1 \times e^{-4}$.

Supervised Fine-Tuning

A 3D UNETR model was trained to segment the seven pulmonary lobes in macaque CT scans using a manually labeled dataset. The UNETR followed the successful U-shaped contracting–expanding pattern but with a ViT encoder to better capture global multi-scale information [13]. The pretrained model weights from the ViT encoder of the ViT autoencoder model were transferred to the ViT encoder in the UNETR. In the UNETR, the ViT encoder was directly connected to the CNN decoder via skip connections at different resolutions. The output was an eight-channel 3D semantic segmentation volume, locating the background and seven pulmonary lobes.

The UNETR semantic segmentation model was trained for 200 epochs on 23 3D CT images with no early termination; training time was approximately 3.5 h on a platform with one CPU core and one GPU. Model weights were saved from the epoch with the highest Dice similarity coefficient (DSC; see below) on the validation dataset. The model was trained with a batch size of one, given GPU constraints. Model parameters were tuned with the Adam optimizer [27] with a fixed learning rate of $1e^{-4}$ and a weight decay of $1e^{-5}$ [13]. The loss function was the DSC (Eq. 4) after applying a sigmoid function.

Quantitative Evaluation

All models were evaluated with sevenfold cross-validation. For each fold, the dataset was split into training-validation-testing sets following a 70:15:15 split. After random shuffling, the

dataset was carefully split such that repeat scans from the same macaque were not spread across sets. Four specific metrics were used to evaluate and report model performance, including DSC, Intersection over Union (IoU), 95% Hausdorff distance (HD_{95}), and average symmetrical surface distance (ASSD). The DSC is a measure of overlap between the prediction and ground truth segmentation:

$$DSC(G, P) = 2 \times \frac{|G \cap P|}{|G| + |P|} \quad (4)$$

where G and P denote the ground truth and prediction images, respectively.

The IoU, or Jaccard index, is another common measure of overlap between the prediction and ground truth segmentation but with a slightly different denominator as compared to DSC:

$$IoU(G, P) = \frac{|G \cap P|}{|G \cup P|} \quad (5)$$

where G and P denote the ground truth and prediction images, respectively.

The Hausdorff distance (HD), or maximum symmetric surface distance, measures the maximum distance between the boundary points of the two masks. The 95th percentile of the distance between the boundary points (HD_{95}) conveys the upper region of the distribution of errors in the boundary points but is not as susceptible to outliers:

$$HD_{95}(G, P) = \max\{d_{95}(S_G, S_P), d_{95}(S_P, S_G)\} \quad (6)$$

where G and P denote the ground truth and prediction images, respectively. S_x denotes the set of surface voxels of mask x , and $d_{95}()$ is the 95th percentile Euclidean distance between boundary points of the ground truth and prediction segmentations.

The ASSD measures the average surface distance error of the symmetrical position between the ground truth and prediction. Compared to the average surface distance, the ASSD better conveys the average error of the boundary points in both directions:

$$ASSD(G, P) = \frac{\sum_{g \in S_G} d(g, S_P) + \sum_{p \in S_P} d(p, S_G)}{|S_G| + |S_P|} \quad (7)$$

where G and P denote the ground truths and output probabilities for all voxels, respectively, S_x denotes the set of surface voxels of mask x , and $d()$ is the Euclidean distance between boundary points of the two segmentations.

Pretraining Dataset Ablation Experiments

To determine the effect of cross-species transfer learning, an ablation experiment was performed in which the

self-supervised training was performed on only one of the two human datasets (Datasets T or H) before supervised fine-tuning was performed. The UNETR model was also trained with random weight initialization to test the no-pretraining condition. These reduced and no human data models were compared to the model pretrained on the full human dataset to evaluate the change in performance. All model hyperparameters for unsupervised pretraining and supervised fine-tuning were kept constant.

Layer Freezing Optimization Experiments

Layer freezing generally refers to the process of locking (freezing) pretrained model weights such that the weights are not updated by the optimizer when fine-tuning a model. The degree of layer freezing has a high impact on the influence of pretraining: too much freezing can inhibit the model's ability to learn the fine-tuning task; in contrast, too little freezing allows the model to completely overwrite pretraining progress. Optimization experiments were performed to determine the optimal degree of layer freezing. The following six transformer encoder layer freezing cases were tested: no freezing, freezing layers 1–3, freezing layers 1–5, freezing layers 1–7, freezing layers 1–9, and freezing all layers (1–12). These cases were compared using the model weights pretrained on the full human dataset (Datasets T and H), and the remaining hyperparameters were kept constant.

Data Augmentation Experiments

Data augmentation was applied before pretraining and fine-tuning to increase the variability of the data, making the model more robust and generalizable. Data augmentation is particularly important for small datasets; however, too much augmentation can inhibit the model's ability to learn the task. To test the effect of our data augmentation on segmentation performance, the full model was compared to models trained with only pretraining augmentations, with only fine-tuning augmentations, and with neither pretraining nor fine-tuning augmentations.

Statistical Analysis

Each model was evaluated using sevenfold cross-validation. The folds were kept consistent across experiments so the performance of each fold could be tracked across different experimental conditions. The model performances were reported as the mean and standard deviation of the DSC, HD_{95} , and ASSD. Wilcoxon signed-rank tests on the test dataset performance were used to statistically identify performance differences across model pretraining conditions.

Results were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) correction [28].

Results

The UNETR segmentation model pretrained using self-supervised learning on 1667 unlabeled human chest CT scans achieved a test performance of $DSC = 90.31 \pm 1.77\%$, $HD_{95} = 8.93 \pm 1.05$ mm, and $ASSD = 1.95 \pm 0.27$ mm. Figure 3 illustrates model performance.

Cross-Species Transfer Learning Performance

To examine the effect of cross-species transfer learning on segmentation performance, an ablation study was performed in which the model pretrained on the full dataset was compared to models pretrained on one of the two datasets and a model that was not pretrained (i.e., randomly initialized weights). Figure 4 illustrates model performance degradation as pretraining data are removed. Table 1 shows the results of the pretraining data ablation experiments for all metrics of interest. The performance improvement was largest when including both Datasets H and T ($\Delta DSC = 1.2\%$, $t = 5.3$, $p = 1.8E-3$), less when training on Dataset H only ($\Delta DSC = 0.7\%$, $t = 3.9$, $p = 8.4E-3$), and least when training on Dataset T only ($\Delta DSC = 0.2\%$, $t = 0.8$, $p = 4.3E-1$).

Per-lobe Performance

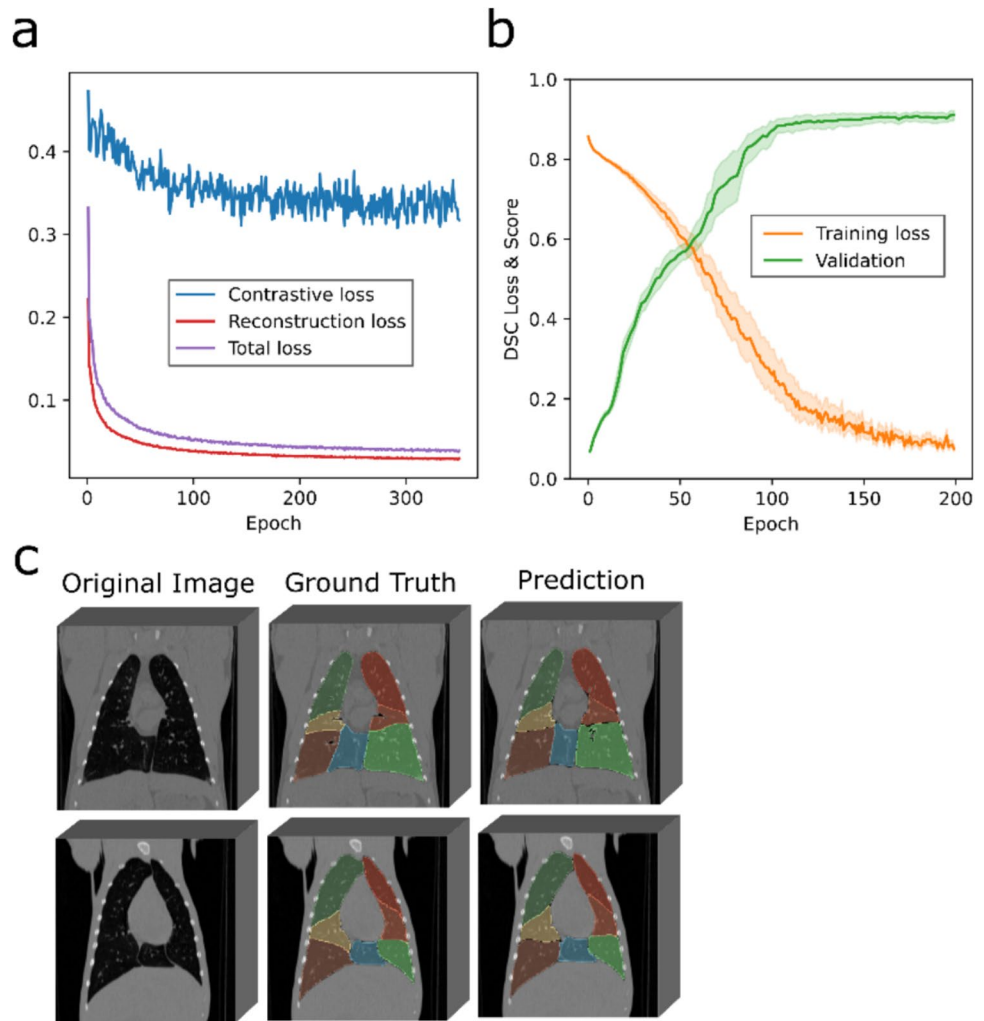
The performance of the segmentation model for each lobe was calculated. The average performance across the seven cross-validation folds is given in Table 2. The model performed better for the upper and lower lobes (average $DSC = 93.12$) compared to the middle and auxiliary lobes (average $DSC = 86.57$). Interestingly, the model performed better for the right (average $DSC = 92.43$) compared to the left lobes (average $DSC = 89.91$), and the lowest performance was observed for the auxiliary lobe.

Layer Freezing Optimization

Optimization experiments were performed to determine the best degree of layer freezing. Starting with the first layer, increasingly more layers were frozen and the model performances were evaluated (Table 3). Overall, freezing layers 1–3 were determined to be optimal and were subsequently used in all other experiments.

Fig. 3 Model performance.

a Self-supervised pretraining learning curves. **b** Supervised fine-tuning learning curves averaged across cross-validation folds. The validation metric was the Dice similarity coefficient (DSC), and the training loss was 1-DSC. The shaded region designates the standard error of the mean. **c** Examples show fully trained pulmonary lobe segmentation performance compared to ground truth



Data Augmentation Effect

We determined the effect of data augmentation on model performance (Table 4). Data augmentation was applied before contrastive learning pretraining and before supervised learning fine-tuning. The same augmentations (random Gaussian noise or smoothing and random elastic transformations) were performed in pretraining and fine-tuning. Segmentation performance was reduced when pretraining and fine-tuning augmentations were both or individually removed. In particular, fine-tuning augmentations highly improved segmentation performance, leading, for example, to a 6% increase in DSC in isolation and a 7% increase in DSC when combined with pretraining augmentations. Overall, these results stress the importance of data augmentation in cross-species transfer learning.

Discussion

In this study, we evaluated human-to-macaque cross-species self-supervised transfer learning as a strategy to enable few-shot learning for pulmonary lobe segmentation in crab-eating macaques. A hand-drawn pulmonary lobe semantic segmentation ground truth dataset was created for a set of 23 macaque chest CT scans, and a dataset of 1667 human chest CT scans was assembled from two sources. Contrastive learning was used to pretrain a 3D ViT on the unlabeled human CT scans. The ViT was used to initialize a 3D UNETR semantic segmentation model that was then trained (fine-tuned) to segment pulmonary lobes from macaque chest CT scans in a supervised manner. We showed that the pretrained model performed highly and significantly better compared either to no pretraining or pretraining using only

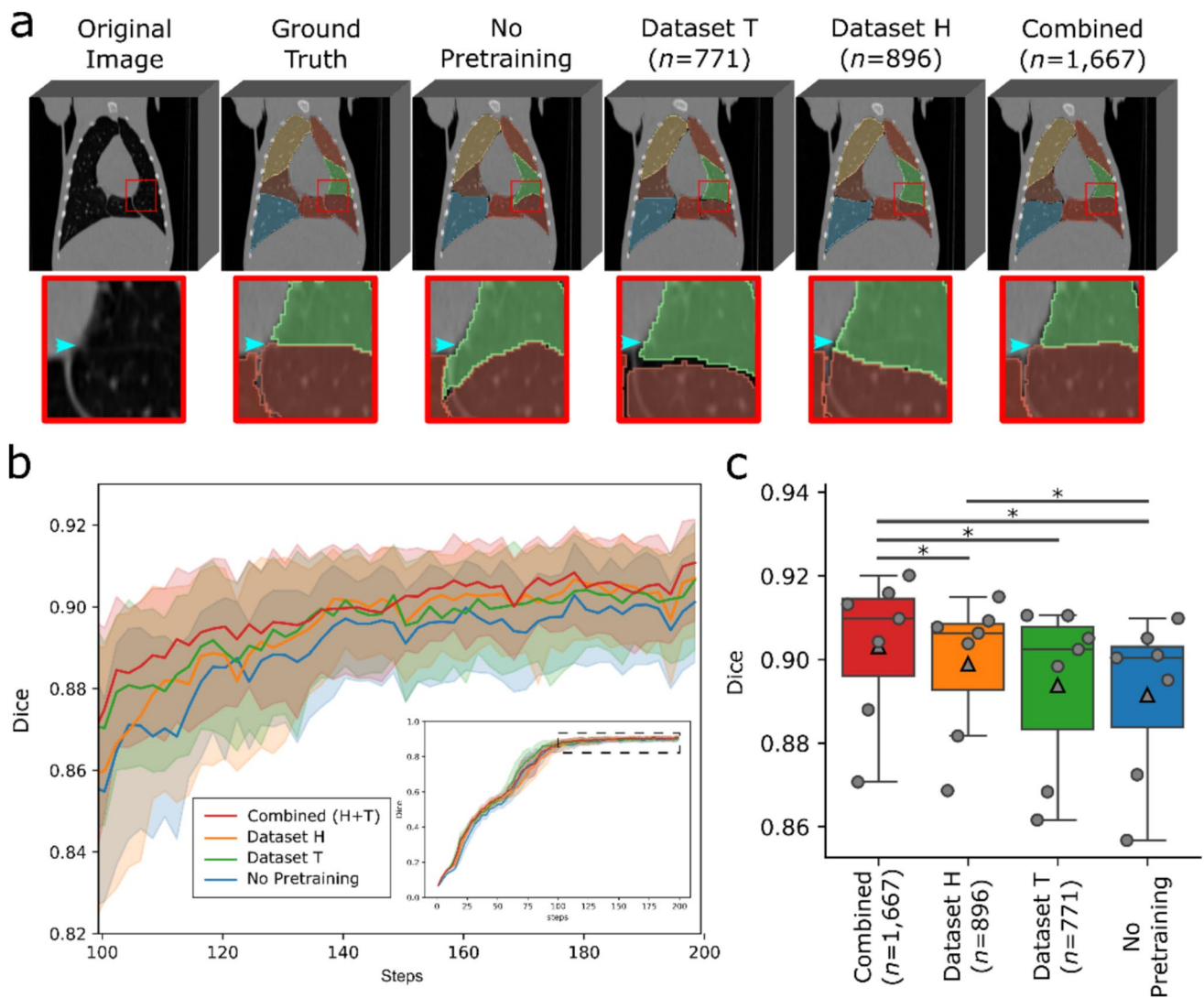


Fig. 4 Pretraining data ablation experiments. **a** Progressive pulmonary lobe segmentation performance degradation was observed as pretraining data was removed. Red boxes show the boundary between the left middle and left lower pulmonary lobe; a light blue arrowhead indicates the fissure delineating the true boundary. **b** Validation curves across four pretraining conditions: no pretraining, Dataset T only, Dataset H only, and combined dataset (T+H). The shaded

region designates the standard error of the mean. **c** Box plots of the test performance across four pretraining conditions; gray dots indicate cross-validation fold performance, and triangles indicate the mean. Wilcoxon signed-rank tests were used to compare the test performance between training conditions, and the p -values were corrected for the false discovery rate using the Benjamini–Hochberg correction. $*p_{\text{fdr}} < 0.05$

Table 1 Ablation on the number of datasets for self-supervised learning pretraining

Human pretraining dataset(s)	DSC (%) $\uparrow$	IoU (%) $\uparrow$	HD ₉₅ (mm) $\downarrow$	ASSD (mm) $\downarrow$
No pretraining	89.15 $\pm$ 1.95	80.99 $\pm$ 2.74	11.60 $\pm$ 2.41	2.22 $\pm$ 0.30
Dataset T (An et al., 2020; $n=771$)	89.38 $\pm$ 2.03	81.39 $\pm$ 2.80	9.93 $\pm$ 1.37	2.16 $\pm$ 0.34
Dataset H (Mostafavi et al., 2021; $n=896$)	89.89 $\pm$ 1.70	82.12 $\pm$ 2.44	9.43 $\pm$ 1.36	2.06 $\pm$ 0.31
Combined (T+H; $n=1667$)	90.31 $\pm$ 1.77	82.79 $\pm$ 2.54	8.93 $\pm$ 1.05	1.95 $\pm$ 0.27

ASSD average symmetrical surface distance, DSC Dice similarity coefficient, HD₉₅ 95% Hausdorff distance, IoU Intersection over Union

The bolded row designates the condition with the highest performance

Table 2 Comparison of segmentation performance among pulmonary lobes

Lobe	DSC (%) ↑	IoU (%) ↑	HD ₉₅ (mm) ↓	ASSD (mm) ↓
Right upper	93.91 ± 1.85	88.74 ± 2.97	8.43 ± 5.08	1.57 ± 0.47
Right mid-dle	89.12 ± 1.88	80.65 ± 2.70	8.60 ± 1.51	2.01 ± 0.24
Right lower	94.26 ± 1.59	89.36 ± 2.39	6.28 ± 0.73	1.59 ± 0.23
Auxiliary	85.18 ± 2.30	74.57 ± 4.25	10.25 ± 2.18	2.21 ± 0.47
Left upper	92.20 ± 1.72	85.70 ± 2.82	7.16 ± 1.20	1.66 ± 0.24
Left mid-dle	85.41 ± 3.59	74.88 ± 5.17	10.69 ± 3.22	2.44 ± 0.60
Left lower	92.11 ± 2.13	85.67 ± 3.34	11.06 ± 2.28	2.17 ± 0.33

ASSD average symmetrical surface distance, DSC Dice similarity coefficient, HD₉₅ 95% Hausdorff distance, IoU Intersection over Union

Table 3 Comparison of layer freezing

Frozen layers	DSC (%) ↑	IoU (%) ↑	HD ₉₅ (mm) ↓	ASSD (mm) ↓
None	89.69 ± 2.25	81.87 ± 3.26	9.93 ± 2.13	2.07 ± 0.35
1–3	90.31 ± 1.77	82.79 ± 2.54	8.93 ± 1.05	1.95 ± 0.24
1–5	90.03 ± 1.91	82.37 ± 2.73	8.75 ± 0.84	1.98 ± 0.30
1–7	89.63 ± 1.96	81.78 ± 2.75	9.61 ± 0.49	2.10 ± 0.25
1–9	90.10 ± 1.78	82.50 ± 2.51	9.05 ± 1.03	1.98 ± 0.26
All (1–12)	89.71 ± 2.07	81.94 ± 2.91	9.42 ± 0.87	2.06 ± 0.29

ASSD average symmetrical surface distance, DSC Dice similarity coefficient, HD₉₅ 95% Hausdorff distance, IoU Intersection over Union

The bolded row designates the condition with the highest performance

one of two human datasets. On a per-lobe basis, model performance was better on upper and lower lobes (compared to middle and auxiliary lobes) and on right lobes (compared to left lobes). Layer freezing optimization experiments revealed that the freezing of transformer layers 1–3 resulted in better

performance than either more or less freezing, albeit by a small margin. Augmentation ablation experiments revealed that data augmentation before pretraining and before fine-tuning had a large positive impact on performance. Our study demonstrated the utility of cross-species self-supervised transfer learning to leverage abundant unlabeled human data to enable few-shot learning for segmentation of medical images from animal models, notably at no additional acquisition or annotation cost.

Previously, the use of labeled human CT scans for tumor segmentation in canines using a U-Net model achieved a 3% improvement in Dice score from transfer learning [7], and the use of labeled macaque CT scans for lung lesion segmentation in humans using a SWIN-UNETR model resulted in a 1% improvement in Dice score [11]. We evaluated the use of unlabeled human CT images for lung lobe segmentation in macaques via a UNETR model and found a 1.2% improvement in Dice score; this performance enhancement is significant, considering that it was achieved using unsupervised learning with publicly available unlabeled data and thus did not require additional acquisition or annotation costs.

Our pretraining data ablation experiments demonstrated a significant performance degradation ($p < 0.05$) when only using one of the two pretraining datasets. Despite similar dataset sizes ($n_H = 896$, $n_T = 771$), the performance improvement from Dataset H (0.74%) was much greater compared to Dataset T (0.23%). Furthermore, the performance boost from using both datasets (1.2%) was greater than the sum of performance improvements from each dataset in isolation (0.97%). In all, these results suggest that the pretraining dataset size and diversity of sources are important for model performance.

We calculated the performance of the segmentation model on a per-lobe basis to evaluate spatial trends in model performance. Improved performance in the segmentation of the upper and lower lobes (versus the middle and auxiliary lobes) is likely related to the lower amount of upper and lower lobe surface area that is along a pulmonary fissure. Fissures are difficult to distinguish on CT images, especially in the presence of pulmonary abnormalities, such as those

Table 4 Comparison of with and without data augmentation

	DSC (%) ↑	IoU (%) ↑	HD ₉₅ (mm) ↓	ASSD (mm) ↓
No augmentations	82.97 ± 2.35	71.92 ± 3.15	17.05 ± 3.01	3.51 ± 0.44
Pretraining augmentations only	84.41 ± 1.63	73.91 ± 2.07	19.78 ± 6.11	3.95 ± 1.54
Fine-tuning augmentations only	89.09 ± 2.17	80.94 ± 3.05	9.71 ± 1.19	2.21 ± 0.33
Pretraining and fine-tuning augmentations	90.31 ± 1.77	82.79 ± 2.54	8.93 ± 1.05	1.95 ± 0.27

Pretraining and fine-tuning augmentations included random Gaussian noise or smoothing and random elastic transformations. Here, “pretraining augmentations” refer to the augmentations applied before contrastive learning augmentations. The bolded row designates the condition with the highest performance.

ASSD average symmetrical surface distance, DSC Dice similarity coefficient, HD₉₅ 95% Hausdorff distance, IoU Intersection over Union

caused by SARS-CoV-2 infection. The right-to-left difference and poor performance in the auxiliary lobe can likely be attributed to recognized anatomical differences between humans (two left and three right lung lobes) and crab-eating macaques (three lung lobes on both sides, plus an auxiliary lobe absent in human lungs).

Layer freezing can have a large impact on the influence of pretraining on the model performance after fine-tuning. Selection of the optimal amount of freezing must balance degrees of constraint and freedom such that the model can adequately adapt to the fine-tuned task but does not overfit to the fine-tuned task. For this task and model design, the optimal amount of layer freezing was determined to be the first three transformer layers, indicating that some early features (i.e., layers 1–3) learned from pretraining (i.e., contrastive learning) were helpful for the fine-tuning task but that deeper features (i.e., layers 4–12) required further training for optimal image segmentation. Future work should consider the degree of similarities between the pretraining (dataset and task) and fine-tuning (dataset and task) when determining the optimal amount of layer freezing.

We acknowledge key limitations of this work that may impact interpretation and generalizability, including a small macaque dataset, the human-to-macaque trajectory, and our use of data from a single virus infection (SARS-CoV-2). A small dataset (in our study, only 23 labeled 3D CT scans of macaques) may lead to a higher risk of overfitting without careful experimental design. With this in mind, we evaluated each model design in a cross-validated manner, ensuring that train-validate-test splits were carefully stratified so as to not split scans from the same subject across datasets. Training a model on a small and labeled dataset (i.e., few-shot learning) is a well-recognized challenge in medical imaging, particularly for laboratories that focus on rare or emerging diseases and animal models of disease. To address this important challenge for the imaging research community, we proposed and validated a strategy utilizing publicly available unlabeled human CT scans to pretrain the model, enabling improved performance on a small labeled dataset without the need for additional data collection or annotation cost.

Our evaluation of human-to-macaque (versus macaque-to-human) pretraining limits the immediate clinical applicability of the work. Nonetheless, our results suggest broader principles regarding cross-species self-supervised transfer learning that are also likely to inform human research. For example, we showed that CT scans from a significantly different primate likely teach the encoder the elements common to human and macaque chest CT images, including intensity distribution, spatial patterns, and variability associated with SARS-CoV-2 exposure. The freezing optimization analysis suggests that the first few layers are likely able to capture these common features. We also demonstrated performance enhancement from the use of self-supervised learning and

ViTs for cross-species transfer learning. Although ViTs have been shown to perform well in image segmentation tasks, they require more training data compared to CNNs, and therefore, cross-species transfer learning might be an effective method to meet the data requirements of ViTs while benefiting from the strengths, such as more efficient learning of global features compared to CNNs.

With regard to generalizability, we specifically developed a pulmonary lobe CT segmentation model for a cohort of SARS-CoV-2–exposed macaques and prioritized pretraining data from humans with COVID-19. SARS-CoV-2 infection is associated with characteristic pulmonary abnormalities on CT scans such as, for example, haziness (i.e., ground-glass opacities) and bright areas (i.e., consolidation). Pretraining on a dataset of human lungs with SARS-CoV-2 infection likely improved the model’s performance on macaque lungs with SARS-CoV-2 infection; however, the performance of cross-species transfer learning may be different for other diseases and should be explored, along with cross-disease transfer learning, in future work.

This study evaluated cross-species self-supervised transfer learning for semantic segmentation, but the techniques are not specific to cross-species or medical imaging and thus have broader applications. Within medical imaging, self-supervised transfer learning would be useful for bridging domains such as modality, site, disease, and demographics. However, as shown in our per-lobe performance analysis, performance is negatively impacted by the magnitude of the differences between the domains and may not be uniform across the image.

Self-supervised transfer learning enhances the target model by exposing it to unlabeled and out-of-domain data that improve its generalizability. In our implementation of self-supervised transfer learning, the number of parameters in the supervised model is unchanged, and therefore, neither the supervised model’s complexity nor the likelihood of overfitting increases. Some of the costs associated with this approach are increased computational resources and implementation time.

Semi-supervised medical image segmentation techniques have been categorized into five (not mutually exclusive) categories: pseudo-labeling, consistency regularization, generative adversarial network (GAN)–based methods, contrastive learning–based methods, and hybrid methods [29]. In pseudo-labeling, the model is trained on the labeled dataset and then applied to the unlabeled dataset to obtain pseudo-labels. The model is then refined on the pseudo-labeled data in an iterative fashion [30]. In consistency regularization, the data and/or model are perturbed to encourage prediction stability [31]. GAN-based methods use a generator and pixel-wise discriminator that classifies pixels as either fake or one of the classes [32]. Contrastive learning–based methods drive positive samples together and negative samples

apart. Contrastive learning–based methods can be subdivided into image-level [22] and patch-level methods [33]. This study focuses on image-based contrastive learning for semi-supervised medical image segmentation but with the added complexity of transfer learning.

Additional experiments could further clarify our findings but were beyond the scope of this study. Obviously, macaque-to-human self-supervised transfer learning would be more translationally impactful but would require a large quantity of unlabeled macaque data and a small quantity of labeled human data, which is not the case in our specific application. Control experiments in which the self-supervised model is trained on 1667 unlabeled macaque (instead of human) CT scans or the supervised model is trained on 1667 labeled human scans and 23 labeled macaque scans might provide comparative insight to separate the effect of cross-species self-supervised transfer learning from the effect of the larger data exposure.

Conclusions

We explored the use of cross-species self-supervised learning of a ViT autoencoder to pretrain the ViT encoder of a UNETR pulmonary lobe segmentation model. To our knowledge, this study is the first to examine cross-species self-supervised transfer learning for image segmentation. We found a significant improvement in segmentation performance from pretraining on chest CT scans from animals of a different species. The improvement in performance increased as the amount of data and number of data sources increased. We showed that the performance of the fine-tuned model was not uniform across the entire lung, possibly due to the differences between human and macaque lung anatomy. We examined the effects of layer freezing and showed that freezing the first 3 out of 12 layers was optimal. We examined the effects of data augmentation and found that data augmentation before pretraining and before fine-tuning had significant positive impacts on performance. These findings provide strong evidence in support of using cross-species self-supervised transfer learning to improve medical image segmentation performance without additional data acquisition or annotation costs. This study paves the way for future work examining cross-species self-supervised transfer learning for segmentation of other animals, organs, and tissues for which labeled data are limited in animals of one species, but unlabeled data are abundant in those of another.

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Data Availability Due to the sensitive nature of nonhuman primate data and the relevant research collaboration agreements, we may not deposit the research data into an online public repository. However, the datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics Approval The experiments were performed in the maximum biosafety level 4 (BSL-4) containment laboratory at the Integrated Research Facility at Fort Detrick (IRF-Frederick). The facility is accredited by the Association for Assessment and Accreditation of Laboratory Animal Care International (AAALAC). The IRF-Frederick is part of the National Institutes of Health (NIH), the National Institute of Allergy and Infectious Diseases (NIAID), and the Division of Clinical Research (DCR). Experimental procedures were approved by the NIAID DCR Animal Care and Use Committee (ACUC) and conducted in compliance with the Animal Welfare Act regulations, Public Health Service policy, and the Guide for the Care and Use of Laboratory Animals (Eighth Edition).

Consent to Participate All human data were anonymized and obtained from previously published datasets/publications.

Consent for Publication All human data were anonymized and obtained from previously published datasets/publications.

Competing Interests The authors declare no competing interests.

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